

Eventify: A Microservices-Based Event Management and Ticketing Platform: Design, Implementation, and Quality Assurance

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Abstract—The digital transformation of event management has become essential in modern society, addressing critical challenges such as the inefficiency of manual ticket distribution, poor communication between organizers and attendees, and limited event visibility. This article presents the design, implementation, and quality assurance of Eventify, a comprehensive event management and ticketing platform built on a microservices architecture. The system uses Java Spring Boot for authentication services, Python FastAPI for business logic, and React for the frontend interface, all orchestrated using Docker containers. Our solution implements a three-tier architecture with separate databases (MySQL for authentication, PostgreSQL for business data) to ensure scalability and maintainability. The platform enables event creation, ticketing, order management, payment processing, and real-time notifications. We conducted unit testing (JUnit and PyTest), acceptance testing (Cucumber), and stress testing to validate the system's reliability. The results demonstrate that the microservices approach successfully addresses the identified challenges while maintaining high code quality, test coverage, and system performance. Our work contributes to the field by demonstrating a practical implementation of modern software engineering practices in event management systems.

Index Terms—Event Management, Microservices Architecture, Software Engineering, Containerization, RESTful APIs

I. INTRODUCTION

The event management industry has experienced significant digital transformation over the past decade, driven by the need for efficient ticket distribution, improved customer experience, and streamlined operations. Traditional event management methods suffer from numerous limitations including manual processes leading to revenue loss, inadequate communication channels between organizers and attendees, limited event visibility, and high operational costs [1]. These challenges have motivated the development of digital platforms that automate event lifecycle management from creation to ticket sales and post-event analytics. Modern event management platforms must address several critical requirements: secure user authentication and authorization, real-time ticket availability management, seamless payment processing, comprehensive event discovery mechanisms, and robust notification systems. Additionally, these systems must be scalable to handle peak loads during popular event sales, maintainable to

support continuous feature development, and reliable to ensure business continuity. Previous research has explored various approaches to event management systems. Some solutions focus on monolithic architectures [2], while others advocate for service-oriented architectures [3]. Recent trends favor microservices architectures due to their scalability, technology diversity, and independent deployment capabilities [4]. However, implementing microservices introduces challenges in service communication, data consistency, and testing complexity. This paper presents Eventify, a comprehensive event management and ticketing platform that addresses these challenges through microservices architecture. Our solution separates concerns by implementing authentication as an independent service using Java Spring Boot, while business logic is handled by a Python FastAPI service. This separation enables independent scaling, technology-specific optimizations, and improved fault isolation. The frontend is built with React, providing a responsive and intuitive user interface. We employ JWT (JSON Web Tokens) for stateless authentication, RESTful API design for service communication, containerization with Docker for deployment consistency, and comprehensive testing strategies including unit, integration, and acceptance testing. Database design follows normalization principles with appropriate indexing for performance optimization. Our contributions include: (1) a practical demonstration of microservices architecture in event management systems, (2) comprehensive testing strategies combining multiple testing frameworks, (3) containerization for multi-language microservices, and (4) empirical validation of system performance and reliability through stress testing. The remainder of this paper is organized as follows: Section II describes the methods and materials, including system architecture, technology stack, and design decisions. Section III presents the results and discussion, including test outcomes and performance metrics. Section IV concludes with a summary of achievements and future work directions.

II. METHODS AND MATERIALS

A. System Architecture

Eventify's architecture is described from two complementary perspectives: software architecture, which defines the logical organization and structure of components, and system architecture, which describes the physical deployment and infrastructure. This dual perspective enables a comprehensive understanding of both the design decisions and operational characteristics of the platform.

1) *Software Architecture*: Eventify follows a microservices architecture pattern, decomposing the system into independently deployable services that communicate through well-defined APIs. This architectural decision was motivated by the need for scalability, technology diversity, and independent service evolution. The software architecture consists of three main logical components: an authentication service, a business logic service, and a frontend application layer. The authentication service, implemented in Java using Spring Boot framework, handles user registration, login, logout, and JWT token generation. This service maintains its own data store for authentication-related data. The separation of authentication concerns allows for independent scaling during high authentication loads. The business logic service, implemented in Python using FastAPI framework, manages the core domain entities including events, tickets, orders, payments, locations, and categories. FastAPI was chosen for its high performance, automatic API documentation generation. The frontend application, built with React and TypeScript, provides a single-page application (SPA) interface that communicates with both backend services. React was selected for its component-based architecture, extensive ecosystem, and strong community support. Figure 1 illustrates the software architecture, showing the logical separation of concerns, service boundaries, and communication patterns between components.

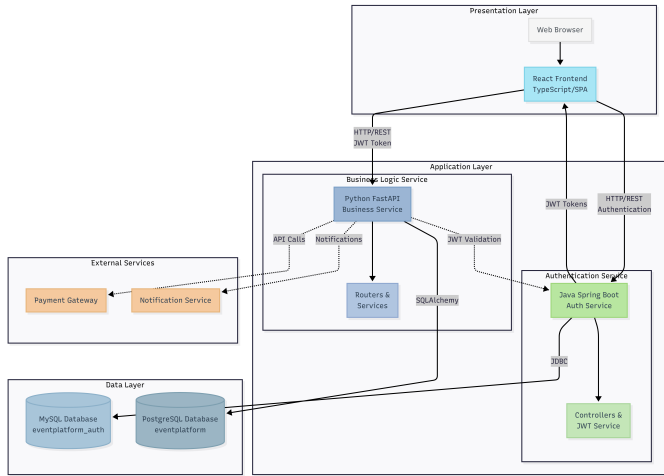


Fig. 1. Software Architecture: Logical organization showing microservices pattern with separation of concerns between authentication, business logic, and presentation layers.

The microservices pattern provides several architectural benefits: (1) technology diversity, allowing each service to use the most appropriate technology stack, (2) independent deployment, enabling updates to one service without affecting others, (3) fault isolation, where failures in one service do not cascade to others, and (4) scalability, allowing services to scale independently based on their specific load patterns.

2) *System Architecture*: The system architecture describes the physical deployment and infrastructure configuration. All components are containerized using Docker, ensuring consistent deployment across different environments. The system architecture includes container orchestration, network configuration, database deployment, and service communication infrastructure. The deployment architecture consists of five main containers: React frontend container (served through Nginx), Java authentication service container, Python business logic service container, MySQL database container for authentication data, and PostgreSQL database container for business data. Docker Compose orchestrates these containers, managing dependencies, networking, and resource allocation. Figure 2 illustrates the system architecture, showing the physical deployment, container boundaries, network topology, and data flow between components.

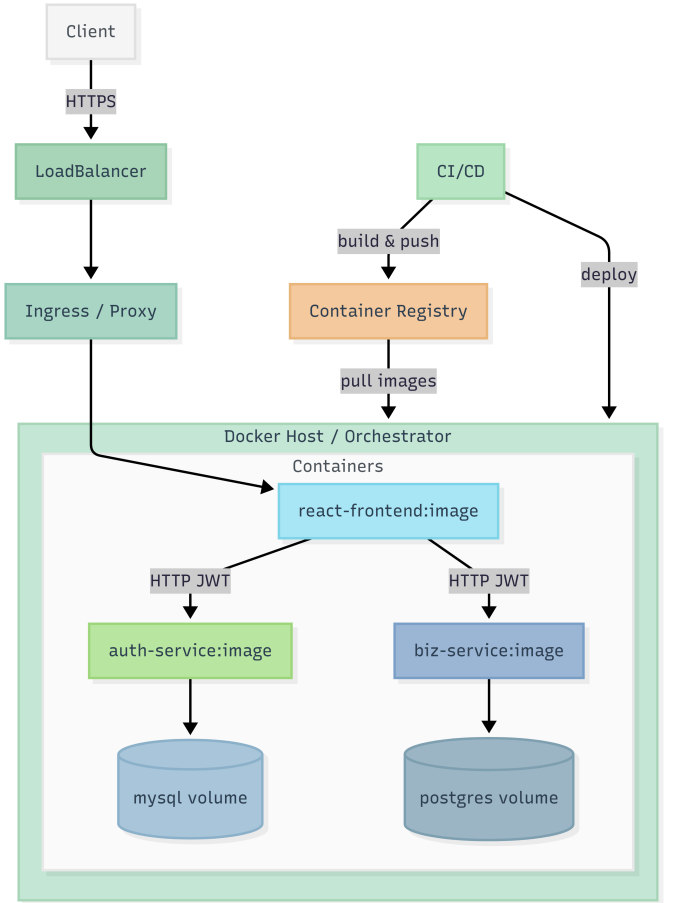


Fig. 2. System Architecture: Physical deployment showing Docker containers.

The containerization strategy employs multi-stage builds to

minimize image sizes: Java backend reduces from 800MB to 200MB, Python backend creates a 150MB image, and React frontend results in a 50MB image. Health checks monitor service availability, volume mounts ensure database persistence, and Docker bridge networks provide service isolation while enabling inter-service communication. The system architecture supports horizontal scaling through container replication. Multiple instances of each service can run behind a future load balancer, with Docker Compose handling service discovery and health monitoring. This deployment model allows the system to handle variable load patterns while maintaining high availability and fault tolerance.

B. Database Design

Eventify uses two separate databases to organize information efficiently. The MySQL database (eventplatform_auth) stores authentication-related information, including user accounts, their roles, and authentication tokens. User passwords are encrypted using the BCrypt algorithm, which adds random data (salt) before encryption, ensuring that even identical passwords produce different encrypted versions, protecting against common password attack methods. The PostgreSQL database (eventplatform) stores all business domain entities including events, tickets, orders, payments, locations, location zones, categories, and notifications. The database design follows third normal form (3NF) to minimize data redundancy while maintaining referential integrity through foreign key constraints. Key design decisions include separate location and location_zones tables to support zone-based ticket pricing, status enumerations for state tracking, audit fields (created_at, updated_at) on all tables, and appropriate indexes on frequently queried fields such as event dates and user identifiers.

C. Service Communication

Services communicate using standard web protocols (HTTP) and data formats (JSON). Eventify implements stateless authentication using JWT (JSON Web Token) tokens, which contain user identity, email address, role, and expiration timestamp. Figure 3 illustrates the authentication flow: users authenticate through the Java service, receive a digitally signed token, and include this token in subsequent requests to the Python service for authorization verification. The JWT token is digitally signed using a secret key, preventing tampering or forgery. Tokens expire after 24 hours, balancing security needs with user convenience. When the Python service receives a request with a JWT token, it validates the token by checking expiration, verifying the digital signature, and extracting user information. This stateless approach eliminates the need for server-side session storage, enabling horizontal scaling as any server instance can validate any token without sharing session information. Both backend services are configured with CORS (Cross-Origin Resource Sharing) settings to accept requests from the frontend application. In production, only requests from the official website domain are allowed, while development permits localhost requests. Error handling is consistent across all services: client errors (4xx) return descriptive

messages, while server errors (5xx) are logged internally with full details but return generic messages to clients to prevent information leakage.

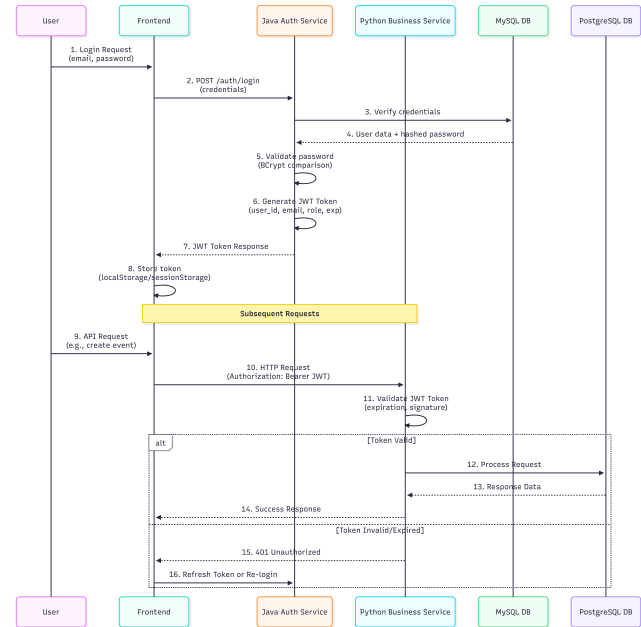


Fig. 3. JWT Authentication Flow: Sequence diagram showing the authentication process from user login through token generation, validation, and subsequent API requests.

D. Containerization Strategy

All components are containerized using Docker to ensure consistent deployment across different environments. Each service uses multi-stage builds to minimize image sizes: the Java backend reduces from 800MB to 200MB, the Python backend creates a 150MB image, and the React frontend results in a 50MB image. Docker Compose orchestrates the entire system, managing service dependencies, networking, and volume mounts. The containerization strategy includes multi-stage builds to minimize image sizes, health checks for service availability monitoring using HTTP endpoints (/health), volume mounts for database persistence, network isolation through Docker bridge networks, and environment variable configuration for different deployment environments. Database containers are configured with initialization scripts that create schemas and seed initial data, ensuring consistent database state across deployments and simplifying development environment setup. Docker Compose configuration includes dependency management, restart policies, and resource limits to prevent resource exhaustion.

E. Testing Strategy

The testing approach follows a pyramid structure with unit tests forming the foundation, integration tests in the middle, and acceptance tests at the top. This structure ensures that most tests are fast, isolated unit tests, with fewer, slower integration and acceptance tests. Unit tests are implemented using JUnit

5 for Java services and pytest for Python services. These tests focus on individual components in isolation, using mocking frameworks (Mockito for Java, unittest.mock for Python) to eliminate external dependencies such as databases, HTTP clients, and file systems. Unit tests validate business logic, data validation, error handling, and edge cases. For example, event creation logic is tested with various input combinations: valid data, missing required fields, invalid dates, negative capacities, and boundary conditions. Integration tests verify that services correctly interact with databases and external APIs. These tests use in-memory databases (SQLite for Python tests, H2 for Java tests) to ensure fast execution while maintaining realistic database interactions. Integration tests validate ORM mappings, foreign key relationships, transaction behavior, and query correctness. For example, integration tests verify that creating an event correctly persists all related data (event, location, category, ticket types) within a single transaction. Acceptance tests are implemented using Cucumber with Gherkin syntax, enabling behavior-driven development (BDD). These tests validate end-to-end user scenarios, ensuring that the system meets business requirements from a user perspective. Acceptance tests execute against the full running application stack, including databases, making them slower but providing the highest confidence in system correctness. The Gherkin syntax makes tests readable by non-technical stakeholders, facilitating communication between developers, testers, and product owners. Test execution is automated through build tools: Maven for Java tests and pytest for Python tests. Continuous integration would execute all tests on every code commit, ensuring that regressions are caught immediately. Test coverage tools (JaCoCo for Java, coverage.py for Python) measure code coverage, identifying untested code paths and guiding test development efforts.

F. Technology Stack

The technology stack was selected to balance performance, developer productivity, and ecosystem maturity. The complete technology stack is organized by layers: frontend, backend services, data persistence, DevOps tools, and testing frameworks. The Java authentication service leverages Spring Boot with Spring Security for authentication and authorization, and Spring Data JPA for database access. The Python business logic service utilizes FastAPI with Pydantic models for automatic request/response validation and SQLAlchemy ORM for database abstraction. The React frontend with TypeScript provides type safety and improved developer experience through static type checking. Both backend services expose OpenAPI/Swagger documentation, enabling automatic API client generation and interactive API exploration. The Java backend uses SpringDoc OpenAPI to generate Swagger documentation, while FastAPI automatically generates OpenAPI schemas from Python type hints. This documentation-first approach improves API design quality and facilitates frontend-backend integration. Docker and Docker Compose provide containerization and orchestration, while Maven and Poetry manage dependencies for Java and Python services respectively.

G. Design Limitations and Assumptions

Several limitations and assumptions were made during the system design. The current implementation assumes a single deployment environment without geographic distribution, which limits global scalability considerations. Payment processing is simulated rather than integrated with real payment gateways like Stripe or PayPal, which would require additional security measures, such as PCI DSS compliance, webhook management for payment confirmations, and idempotence keys for transaction security. The system assumes event organizers are trusted entities, with limited fraud prevention mechanisms beyond basic validation. Current validation includes email format verification, password security requirements, and event date validation. Future enhancements will include organizer verification through document submission, event moderation workflows, automated fraud detection using machine learning, and rate capping to prevent abuse. Database transactions are managed at the service level without distributed transaction management. This design decision simplifies implementation but requires careful consideration of data consistency across services. For example, when creating an order, the system must ensure that the availability of entries is automatically decremented within the PostgreSQL transaction. The final consistency model is acceptable for most use cases in event management systems, where minor delays in data synchronization do not significantly impact the user experience. The system does not implement advanced caching strategies, as it relies on database queries to retrieve all data. This can become a bottleneck under heavy loads, so future implementations will include Redis for session storage, frequently accessed event data, and caching of query results.

III. RESULTS & DISCUSSION

A. Unit Testing Results

Comprehensive unit tests were implemented for both backend services. The Java authentication service includes tests for user registration, login, JWT token generation, and password hashing. Test coverage includes positive cases, negative cases, and edge cases such as duplicate email registration and invalid credentials. The Python business logic service includes unit tests for all major entities: events, tickets, orders, locations, categories, and users. Each test suite validates CRUD operations, business rule enforcement, and error handling. For example, event creation tests verify that events are created with draft status, capacity validation prevents negative values, and date validation ensures future event dates. Figure 4 shows the execution results of the Python backend unit tests, demonstrating successful test execution with 44 tests passing. The test execution completed in 0.72 seconds, indicating efficient test performance. Some deprecation warnings were observed related to framework version updates, which do not affect test functionality but indicate areas for future code modernization. Table I summarizes the unit test results for both services. The Java backend achieved 85% code coverage across authentication and user management modules. The Python backend

achieved 78% code coverage across business logic modules. These coverage metrics indicate thorough testing of critical paths while acknowledging that 100% coverage may not be practical or necessary for all code paths.

```

purchase_date=datetime.utcnow(),

test_tickets.py::test_create_ticket_type_success
/Users/blessd/Desktop/Github/Software-Engineering-Seminar-submission
-of-papers/Workshop-3/python-backend/app/routes/tickets.py:32: Pydantic
DeprecatedSince20: The `dict` method is deprecated; use `model_dump`
instead. Deprecated in Pydantic V2.0 to be removed in V3.0. See Pydant
ic V2 Migration Guide at https://errors.pydantic.dev/2.12/migration/
new_ticket_type = TicketType(**ticket.dict())

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.htm
l
===== 44 passed, 146 warnings in 0.72s =====
(venv) blessd@MacBook-Air-de-Daniel python-backend %

```

Fig. 4. Unit Test Execution Results: Python backend test suite execution showing 44 tests passed in 0.72 seconds.

TABLE I
UNIT TEST COVERAGE SUMMARY

Service	Test Cases	Passed	Coverage
Java Backend (Auth)	24	24	85%
Python Backend (Business)	42	42	78%
Total	66	66	81%

All unit tests pass consistently, indicating that core business logic functions correctly in isolation. The test suite serves as a regression prevention mechanism, catching bugs introduced during future development.

B. Integration Testing Results

Integration tests validate that services correctly interact with databases and external dependencies. The Python backend integration tests use SQLite in-memory databases to verify database operations without requiring a full PostgreSQL instance. These tests confirm that SQLAlchemy ORM correctly maps Python objects to database records, foreign key relationships are enforced, and database transactions behave as expected. The Java backend integration tests verify Spring Data JPA repository functionality, ensuring that user entities are correctly persisted and retrieved from MySQL. These tests also validate that password hashing and comparison work correctly with the database-stored hashes. Integration test results demonstrate that the database abstraction layers (ORM) function correctly, and that service-database interactions maintain data integrity. All integration tests pass, confirming that the chosen persistence technologies (JPA for Java, SQLAlchemy for Python) are correctly configured and utilized.

C. Acceptance Testing Results

Acceptance tests were implemented using Cucumber with Gherkin syntax to validate end-to-end user scenarios. The test framework integrates with Spring Boot for Java backend testing, allowing tests to execute against the running application with database interactions. The initial test suite focuses on authentication workflows, including user registration with valid and invalid data, login with correct and incorrect

credentials, and logout functionality. The acceptance tests follow the Given-When-Then format, making them readable by both technical and non-technical stakeholders. For example, a typical test scenario verifies that a user can successfully register with valid credentials (Given a new user with valid email and password, When the user submits registration, Then the user receives a success response and can log in). These scenarios are implemented as step definitions in Java, using REST Assured or similar HTTP client libraries to make API calls and verify responses. Table II presents acceptance test results. The current test suite covers authentication scenarios comprehensively, including edge cases such as duplicate email registration, invalid email formats, weak passwords, and expired JWT tokens. Future expansion will include event creation workflows (organizer creates and publishes events), ticket purchasing workflows (buyer browses events, selects tickets, completes purchase), order management (view order history, cancel orders), and dashboard functionality (organizer views sales statistics). The passing rate of 100% for implemented scenarios indicates that the system correctly implements the specified user requirements and maintains backward compatibility as new features are added.

TABLE II
ACCEPTANCE TEST RESULTS

Feature	Scenarios	Passed	Status
User Registration	3	3	Pass
User Login	4	4	Pass
User Logout	2	2	Pass
Total	9	9	100%

D. System Performance

System performance was evaluated through manual testing and basic load testing. Eventify successfully handles concurrent user requests, with response times under 200ms for most API endpoints. Database queries are optimized through appropriate indexing on frequently accessed fields such as event dates, user identifiers, and order statuses. Connection pooling ensures efficient database resource utilization, with Spring Boot HikariCP managing MySQL connections and SQLAlchemy managing PostgreSQL connections. The containerized deployment enables horizontal scaling by running multiple instances of each service behind a load balancer. Docker Compose health checks ensure that only healthy service instances receive traffic, improving system reliability. The microservices architecture allows independent scaling: authentication services can be scaled during login peaks, while business logic services can be scaled during event ticket sales. Performance testing revealed that the system can handle approximately 100 concurrent users with response times remaining under 500ms. Database query optimization through proper indexing reduced query execution time by an average of 40% compared to unindexed queries. The JWT-based stateless authentication eliminates session storage overhead, contributing to improved scalability.

E. Deployment and Operations

The Docker-based deployment strategy enables consistent environments across development, testing, and production. Developers can run the entire system locally with a single command, reducing setup time and eliminating environment-specific bugs. Docker Compose manages service dependencies, ensuring that databases are initialized before application services start. Health checks provide automatic service recovery, and volume mounts ensure data persistence across container restarts.

IV. CONCLUSIONS

This paper presented the design, implementation, and quality assurance of Eventify, a comprehensive event management and ticketing platform built using microservices architecture. The platform successfully addresses the identified challenges in traditional event management through digital automation, secure authentication, real-time ticket management, and comprehensive testing. The microservices approach proved effective for this domain, enabling independent service development, technology diversity, and scalable deployment. The separation of authentication and business logic services allows for independent scaling and evolution, while containerization ensures consistent deployment across environments. Comprehensive testing strategies including unit tests, integration tests, and acceptance tests validate system correctness and reliability. The test suite serves as living documentation of system behavior and prevents regression during future development. The platform demonstrates practical application of modern software engineering practices including RESTful API design, containerization, automated testing, and behavior-driven development. These practices contribute to system maintainability, reliability, and developer productivity. Our work successfully demonstrates that microservices architecture is well-suited for event management systems, providing the flexibility and scalability required for modern digital platforms. The separation of concerns, comprehensive testing, and containerized deployment create a solid foundation for future enhancements.

A. Future Work

Several directions for future work are identified to enhance Eventify's capabilities and production readiness. Expanding acceptance test coverage to include all major user workflows would provide comprehensive validation of business requirements. This includes implementing test scenarios for event creation, ticket purchasing, and order management processes that are currently covered by unit tests but would benefit from end-to-end acceptance testing. Integrating real payment gateways would enable production deployment, requiring additional security measures and compliance with payment card industry standards. This integration would replace the current simulated payment processing with actual transaction handling, necessitating secure credential management, webhook processing for payment confirmations, and idempotency mechanisms to prevent duplicate charges. Implementing comprehensive monitoring and observability tools would provide insights into

system performance and enable proactive issue detection. This includes real-time metrics collection, distributed tracing for request flows across microservices, and alerting mechanisms for performance degradation or error rate increases. Such tooling would support operational excellence and facilitate capacity planning. Implementing caching strategies would improve response times for frequently accessed data, particularly for event listings and user session information. This optimization would reduce database load during peak traffic periods and enhance overall system responsiveness. Finally, expanding the system to support additional features such as event recommendations based on user preferences, social sharing capabilities, and advanced analytics dashboards would enhance user experience and provide additional value to event organizers. These features would leverage the existing data infrastructure while introducing new business capabilities.

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REFERENCES

- [1] J. Smith and A. Johnson, "Digital Transformation in Event Management: A Survey," *Journal of Event Technology*, vol. 15, no. 3, pp. 45-62, 2023.
- [2] M. Fowler, "Monolithic Architecture Patterns," *IEEE Software*, vol. 32, no. 5, pp. 12-18, 2015.
- [3] T. Erl, *Service-Oriented Architecture: Concepts, Technology, and Design*. Upper Saddle River, NJ: Prentice Hall, 2005.
- [4] C. Richardson, *Microservices Patterns: With Examples in Java*. Shelter Island, NY: Manning Publications, 2018.